2D Vector Navigation Math: Spin-then-Drive Model

The system takes a 2D Cartesian coordinate vector ($v = X\hat{i} + Y\hat{j}$) and calculates the required physical maneuvers. The rover is physically placed facing the positive X-axis ($0°$).

Phase 1: Calculus Vector Math

- **Magnitude (Distance):** $M = \sqrt{(X^2 + Y^2)}$
- **Target Angle (Heading):** $\theta = \text{atan2}(Y, X) \times (180 / \pi)$

Phase 2: Translation to Hardware Timers

The mathematical values are converted into physical motor run-times in milliseconds using calibration constants tuned to the floor's friction.

- **Spin Duration:** $\text{Spin Duration} = |\theta| \times \text{SPIN_MS_PER_DEGREE}$
- **Drive Duration:** $\text{Drive Duration} = M \times \text{DRIVE_MS_PER_UNIT}$

Phase 3: Physical Execution

The rover uses a state machine to evaluate the target angle and determine the direction of rotation.

- **Positive Angle:** The rover spins Left (Counter-Clockwise).
- **Negative Angle:** The rover spins Right (Clockwise).
- **Zero Angle:** The rover skips the spin phase and immediately drives forward.